



Splunk: A SIEM Tool – Bootcamp Notes

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01. What is Splunk?

Splunk is a **data analytics and monitoring platform** that collects and analyzes **machine-generated data** (like logs from servers, networks, or applications) to find patterns and gain useful insights.

Key Uses of Splunk

1. **Data Analytics** – Helps discover patterns in big data for decision-making.
 2. **Cybersecurity** – Detects threats and anomalies in real time from system logs.
 3. **Operational Intelligence** – Improves IT and business operations by identifying issues early.
 4. **Log Management** – Centralizes logs from different systems for faster troubleshooting.
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Why Use Splunk?

- **Real-Time Insights:** Instant visibility for quick decisions.
 - **Scalability:** Handles large data volumes easily.
 - **Powerful Search:** Finds hidden data patterns quickly.
 - **Data Correlation:** Combines data from many sources.
 - **Custom Dashboards:** Visualize key metrics in one place.
 - **User-Friendly Interface:** Easy to navigate.
 - **Strong Security:** Built-in monitoring and threat detection.
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Before Splunk

- **Problem:** Data was scattered across silos, making management difficult.
- **Solution:** A centralized system for managing and analyzing all data.
- **Result:** Streamlined operations and better access to information.

Implementation Approach:

1. **Identify** issues in current data handling.
 2. **Evaluate** existing systems.
 3. **Plan** integration strategies.
 4. **Implement** centralized tools (like Splunk).
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Pros and Cons of Splunk

Pros

- Turns large data into useful insights.
- Provides real-time monitoring and alerts.

Cons

- Expensive for big data volumes.
 - Complex setup and configuration.
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Motivation for Using Splunk

- **Challenge 1:** Hard to combine data from multiple sources → slow decisions.
 - **Challenge 2:** Needed better security → regulatory compliance and auditing.
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Splunk in Cybersecurity

- **Problem:** Slow threat detection.
 - **Solution:** Real-time monitoring and alerting.
 - **Benefit:** Faster incident response and stronger defense.
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Network Analysis with Splunk

- **Traffic Insights:** Monitors network traffic and detects unusual activity.
 - **Incident Response:** Analyzes logs to quickly respond to threats.
 - **Performance Metrics:** Tracks network performance (bandwidth, latency, etc.) to prevent slowdowns.
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In Short

Splunk is a **powerful platform** for analyzing and visualizing data from many systems in real time. It's especially valuable in **cybersecurity**, **network monitoring**, and **operational management**, helping organizations make faster, data-driven decisions.

02. Splunk Architecture and Setup

Splunk works through **three main stages** and **three main components**.

1. Data Input Stage

- Splunk collects raw data from sources (like servers, firewalls, applications, etc.).
- It divides data into **64KB blocks** and adds **metadata** (like hostname, source, and sourcetype).
- This helps Splunk understand where data came from and how to process it later.

Think of this as “collecting and tagging” data before analysis.

2. Data Storage Stage

This has two sub-stages: **Parsing** and **Indexing**.

◆ Parsing Phase

- Splunk breaks the data stream into **individual events** (lines).
- Adds timestamps and metadata.
- Removes unnecessary data using **regular expressions (regex)**.

◆ Indexing Phase

- The processed events are saved to **indexes** on disk.
- Both raw and compressed data are stored.
- Makes searching and visualization **much faster**.

This is where Splunk converts data into searchable form.

3. Data Searching Stage

- This is where users **search, view, and analyze** indexed data.
- It includes **dashboards, alerts, and reports**.
- Splunk creates **knowledge objects** (saved searches, event types, field extractions).

You use this stage to visualize insights and detect issues.

Main Splunk Components

1. Splunk Forwarder

Collects and sends data to the indexer.

- **Universal Forwarder:** Sends raw data with minimal processing.
- **Heavy Forwarder:** Parses and filters data before sending (saves bandwidth).

♦ *Used in remote servers to forward logs automatically.*

2. Splunk Indexer

- Receives data from forwarders.
- **Parses and stores** events into indexes.
- Makes data **searchable** and **query-ready**.

♦ *Acts like a database where all your logs are stored and optimized.*

3. Splunk Search Head

- User interface for searching and creating dashboards.
- Allows users to run queries and view reports.

♦ *What you see when you log into Splunk via a browser.*

Additional Features

- Splunk can **collect logs from multiple machines** using network ports.
 - Can **clone, load balance**, and **route** data intelligently.
 - Uses a **Deployment Server** to manage all configurations centrally.
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Installation Process

For Linux

1. Download .deb package from: splunk.com/download
2. Go to the downloaded directory and install it.
3. Start Splunk using the command with license acceptance.
4. Create an **admin username & password**.
5. Access via: `http://localhost:8000`

For Windows

1. Download the .msi installer.
 2. Run the installer and choose **Local System**.
 3. Set an admin password and finish setup.
 4. Access via browser → `http://localhost:8000`
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Splunk Licensing Overview

Splunk licenses depend on **how much data you index per day**.

Types of Licenses:

1. **Enterprise License** – Full features (authentication, distributed search).
 2. **Free License** – Limited indexing; fewer features.
 3. **Forwarder License** – Only for forwarding data.
 4. **Beta License** – For test versions.
 5. **Premium App License** – Adds extra features to Enterprise/Cloud.
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In Short

Stage	Function	Component
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Input	Collect raw data	Forwarder
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Storage	Parse & index data	Indexer
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Search	Query & visualize	Search Head
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Splunk transforms **machine data** → **searchable insights**, making it essential for **cybersecurity, monitoring, and data analytics**.

03. Splunk Searching (Mastering Data Insights)

Splunk allows you to search and analyze large sets of data quickly.

Efficient searching helps you get **faster and more accurate insights** from logs and events.

1. Splunk Search Processing Language (SPL)

SPL (Search Processing Language) is the **core language** for searching in Splunk.

It includes **commands, functions, and operators** to query and filter data.

Basic Syntax Elements:

- search – keyword to find specific data.
- index – specifies which index (data location) to search.
 - Example: index=main
- sourcetype – filters data by source type.
 - Example: sourcetype=apache_access
- host – filters events by host machine.
- Logical operators: AND, OR, NOT – used to combine or exclude conditions.

Example:

```
index=web sourcetype=nginx_access status=200 OR status=404
```

2. Filtering and Extracting Data

Splunk can refine searches using **commands** that process data in stages.

Command Function		Example Use
where	Filter events by conditions	where status_code>=400
rex	Extract fields using regex (pattern matching)	rex field=_raw "user=(?<username>\w+)"
extract	Pull fields based on key-value pairs	extract pairdelim="," kvdelim="="

● Purpose:

- **Refine:** Narrow down search results.
 - **Extract:** Get meaningful fields (like usernames, IPs).
 - **Modify:** Adjust data values as needed.
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3. Transforming Data into Actionable Reports

Once data is filtered, you can **summarize and visualize** it.

Command	Description	Example
stats	Summarize data (count, avg, sum, etc.)	stats count by status_code
timechart	Create time-based graphs	timechart avg(response_time)
table	Show results in a simple table	table host, status_code, response_time
fields	Select which fields to display	fields host, user, method

Process:

1. **Aggregate** – Summarize data for insights.
 2. **Analyze** – Spot trends and patterns.
 3. **Visualize** – Create charts and dashboards.
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4. Advanced Searching Techniques

Splunk allows **dynamic and enriched searches** with these features:

Feature	Description	Example Use
Subsearches	A search inside another search for filtering	<code>`index=web [search error</code>
Lookups	Enrich data using external sources (CSV, DB)	Maps user IDs to usernames
Transaction	Groups related events together	Track user login → logout sessions

These tools make searches more powerful and context-aware.

5. Real-Time Web Server Monitoring Example

Monitor your web server logs (like Apache/Nginx) using Splunk.

Goal: Detect

- Error spikes (e.g., 500 errors)
- Slow response times
- Suspicious/malicious activity

Example search:

```
index=web sourcetype=apache_access status_code>=500
```

Steps:

1. **Collect:** Get web logs from Apache/Nginx.
 2. **Analyze:** Look for high status_code values.
 3. **Identify:** Detect spikes and patterns (e.g., attacks or performance issues).
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In Short

Splunk's SPL helps you:

- Filter and refine large data sets efficiently.
- Aggregate and visualize results through dashboards.
- Perform **real-time monitoring** for security and performance.

Remember:

- search, stats, timechart, and rex are core commands.
- Combine **basic + advanced** techniques for deeper insights.

04. Splunk Alerting and Reporting for Real-Time IT Monitoring

Splunk helps organizations **monitor systems in real time**, detect issues early, and make data-driven decisions — used by **90+ of the Fortune 100** companies.

1. Core Concept: Splunk Alerts

Alerts automatically notify users about important or abnormal events in data.

Types of Alerts

Type	Description	Example
Scheduled Alert	Runs at fixed time intervals.	Daily error summary.
Real-Time Alert	Triggers instantly when a condition is met.	Detects a security breach.
Triggered Alert	Activated when specific criteria are true.	CPU > 90% usage.
Alert Actions – Can send emails , run scripts , or call external systems.		

2. Steps to Configure an Alert (via GUI)

1. **Define Query** – Write the SPL search query.
 2. **Set Conditions** – Specify what event should trigger the alert.
 3. **Throttle** – Add time gaps to avoid repeated notifications (reduces alert fatigue by 25%).
 4. **Define Actions** – Choose how Splunk will notify (email, script, etc.).
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3. Real-Time Alert Examples

Example 1 – Security Breach Detection

- Detect logins from unusual locations.
- Instant alerts notify the security team.
- Prevents financial loss (avg. breach cost ≈ \$4.45 M).

Steps: Identify Threat → Instant Alert → Secure System

Example 2 – Application Performance Monitoring

- Monitors web response time.
- Alerts when response > threshold.
- Prevents slow performance (40% of users leave slow sites).

Steps: Track Response → Identify Slow Spots → Alert Operations

4. Core Concept: Splunk Reports

Reports turn raw data into **clear visual insights** for better decision-making.

Types of Reports

Type	Description
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Scheduled Reports	Generated at fixed intervals automatically.
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Summary Reports	Display aggregated summaries (e.g., daily activity).
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Dashboards	Visual interactive reports for real-time monitoring.
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5. Creating Reports – Practical Steps

1. **Search Query:** Define what data to analyze.
2. **Time Range:** Set duration (e.g., last 24 hours).
3. **Visualization:** Pick chart types (bar, line, pie) for clarity.

Dashboards = for executives

Scheduled reports = for technical teams

6. Real-Time IT Example – Capacity Planning

- Monitor disk space usage and predict future capacity.
- Identify resource bottlenecks early.
- Predictive reports improve planning by ≈ 15%.

Steps: Track Usage → Predict Needs → Visualize Trends

7. Best Practices for Alerts & Reports

Practice	Purpose
Throttle Alerts	Avoid repeated notifications.
Use Clear Names	Easier management and review.
Regular Review	Ensure alerts are still relevant.
Document Configs	Keep track of settings.
Centralize Management	Maintain better control over multiple alerts.

8. Resources for Learning More

- **Splunk Docs** – Official documentation.
 - **Community Forums** – Peer discussions and troubleshooting.
 - **Enterprise Security & ITSI** – Advanced analytics apps.
 - **Splunkbase** – Download pre-built apps and dashboards.
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In Short

Splunk Alerting and Reporting helps organizations:

- Detect problems instantly.
- Keep systems secure and optimized.
- Provide clear, real-time insights for operations and management.

05.Splunk Dashboards: Turning Data into Insights

Splunk dashboards help convert **raw machine data** into **actionable, real-time insights** using powerful visual tools.

They're vital for **monitoring cybersecurity, analyzing networks, and supporting business decisions**.

1. Powerful Dashboard Features

Feature	Description	Benefit
Real-Time Monitoring	Dashboards update instantly with live data streams.	Detect issues as they happen — no delays.
Rich Visualizations	Use charts, graphs, and tables to make complex data clear.	Quickly identify patterns and trends.
Customizable Layouts	Use Dashboard Studio to drag and drop elements.	Tailor dashboards to specific teams or goals.

Dashboards simplify data interpretation and speed up decision-making.

2. Creating Your First Dashboard

1. **Define Your Goal** – Choose what you want to monitor (e.g., login failures, response times).
 2. **Write SPL Queries** – Use Splunk Processing Language to extract key data.
 3. **Select Visualizations** – Pick chart types that best show the data:
 - Pie chart → proportions
 - Line chart → trends
 - Bar chart → comparisons
 4. **Save & Share** – Publish dashboards for teams and set **auto-refresh** intervals.
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3. Cybersecurity Monitoring Use Case

Step	Action	Purpose
Detect	Dashboard flags unusual logins or “impossible travel” (same user logging from two countries quickly).	Identifies suspicious activity.
Analyze	Drill-down features let analysts inspect events and correlate with threat data.	Investigate threats effectively.
Respond	Automated alerts and workflows trigger incident response.	Contain and resolve security incidents in real-time.

Dashboards make threat detection and response faster and smarter.

4. Network Analysis in Action

Function	Description	Value
Traffic Visualization	Heatmaps show bottlenecks using color-coded severity.	Instantly locate network slowdowns.
Performance Metrics	Monitor latency, packet loss, and throughput.	Compare current vs. historical data.
Capacity Planning	Use trend analysis to predict bandwidth needs.	Prevent outages and optimize costs.

5. Why Learn Splunk Now?

Metric Insight

94% Fortune 100 companies use Splunk.

\$130K+ Average US salary for Splunk-certified professionals.

47% Year-over-year job growth for Splunk-related roles.

Splunk skills offer strong job security and career advancement.

6. Advanced Career Pathways

Role	Focus
Splunk Architect	Design and implement Splunk across enterprises.
Security Analyst	Use Splunk as a SIEM tool for threat detection.
Data Scientist	Apply Splunk ML features for analytics.
Administrator	Manage and maintain Splunk systems.
Developer	Build custom Splunk apps and extensions.

7. Driving Business Value with Splunk

Benefit	Description
Enhanced Security	Detect and prevent attacks proactively.
Operational Efficiency	Reduce Mean Time To Resolve (MTTR) by up to 70%.
Data-Driven Decisions	Make smarter choices based on real evidence.
Competitive Advantage	Act faster than competitors using real-time insights.

In Short

Splunk Dashboards:

- Turn data into live, visual insights.
- Improve security, performance, and operations.
- Are valuable skills for IT, cybersecurity, and data careers.

Mastering Splunk dashboards boosts both your organization's intelligence and your professional growth.